

Zero-Shot Large Audio-Language Models versus Trained Captioners: A Preregistered Evaluation on Clotho with Polyphony and Hallucination Analyses

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Abstract—Large audio-language models (LALMs) promise open-vocabulary audio understanding without task-specific training, but whether they match trained audio captioners remains unclear. We evaluate three zero-shot LALMs (Qwen2.5-Omni-7B, SALMONN-13B, Audio Flamingo 3) against three traditional systems (an AST tagging floor, the DCASE 2023 CNN14+BART baseline, EnCLAP-base) on the full Clotho v2.1 evaluation split under one harness with identical data, decoding provenance, and DCASE metrics. The CNN14 baseline reproduces its published score within 0.002 SPIDER-FL, validating the harness; the EnCLAP row scores ≈ 0.011 below its published value, which we disclose. Zero-shot performance is strongly model-dependent: two LALMs trail both trained captioners, while Audio Flamingo 3 surpasses them on every metric in-harness (SPIDER-FL 0.297 vs. 0.280, also above EnCLAP’s published 0.291; bootstrap CI excludes the preregistered 0.261 threshold). A PANNs-based polyphony split shows *higher* caption-overlap scores on event-rich polyphonic clips for all six systems, and a closed-vocabulary hallucination metric (CHAIR-audio) shows the best captioner does not hallucinate less than its LALM peer. “Zero-shot” denotes the evaluation protocol (no fine-tuning, no in-context examples); the audio-specialist LALMs’ training corpora include Clotho development pairs, which we disclose and discuss. All hypotheses were preregistered; deviations are disclosed. Code: <https://github.com/Zuraiz270/audio-to-text-captioning-using-LALMs>.

I. INTRODUCTION

Automated audio captioning (AAC) maps an audio recording to a natural-language description of its sound events. Traditional AAC systems are trained end-to-end on caption corpora such as Clotho [1]; large audio-language models (LALMs) instead couple a pretrained audio encoder to a large language model and are used *zero-shot*. Whether zero-shot LALMs already match purpose-trained captioners — and where they fail — is the question this project answers empirically. We ask: **RQ1**: Do zero-shot LALMs match or exceed traditional captioning baselines on standard metrics (SPIDER-FL, CIDEr)? **RQ2**: How does performance differ between polyphonic (overlapping events) and monophonic clips? **RQ3**: What are the failure modes, in particular caption hallucination?

Our contributions are: (i) a reproducible six-system comparison on the full Clotho-eval split, anchored by baselines

that reproduce their published scores; (ii) a preregistered, bootstrap-based hypothesis-testing protocol with all deviations disclosed; (iii) an SED-derived polyphony split and a closed-vocabulary adaptation of the CHAIR hallucination metric [2] to audio.

II. RELATED WORK

Clotho v2.1 [1] is the standard AAC benchmark: 1,045 evaluation clips (15–30 s, 44.1 kHz) with five human captions each. The DCASE challenge series established SPIDER [3] and its fluency-penalised variant SPIDER-FL [4] as headline metrics; SPIDER-FL’s fluency detector also underlies FENSE, the caption metric used in [5]. We compute all metrics with `aac-metrics` [6]. Trained captioners include the DCASE 2023 baseline (CNN14+BART, encoder from PANNs [7]) [8] and EnCLAP [9]. AST [10] is a 527-class AudioSet [11] tagger we use as a tagging floor. The LALMs are Qwen2.5-Omni [12], SALMONN [13], and Audio Flamingo 3 (AF3) [14]. For polyphonic scene understanding, Abeßer et al. [15] introduced the USM dataset, and Harish and Abeßer [5] showed LALM performance on *event-level* tasks degrades with increasing polyphony — a finding our *caption-level* RQ2 complements. MACE [16] scores captions against the audio itself via CLAP embeddings; we treat it as a secondary metric.

III. METHOD

A. Evaluation Harness

Every system implements one contract — `caption(waveform, sr) → str` — behind a registry, so a single inference loop and a single scorer serve all rows. Each run writes a predictions JSON plus a manifest recording checkpoint SHA-256s, library versions, decode parameters, and the seed (42). Scoring uses the `dcase2023` preset of `aac-metrics` 0.5.5 (SPIDER-FL, CIDEr-D, SPICE, METEOR, fluency error) on identical clip sets; a `-subset` mechanism restricts scoring to fixed clip lists for same-set comparisons. Baselines ran on CPU; LALMs ran on one NVIDIA A100-40GB each (offline, pre-cached checkpoints).

B. Systems

AST (tagging floor): top-5 sigmoid AudioSet labels templated as “a sound of ...”; input window 10.24 s. **CNN14+BART**: the official DCASE 2023 baseline, reproduced at 0.259 SPIDeR-FL vs. the published 0.261 — this reproduction validates the harness. **EnCLAP-base** (AudioCaps-pretrained, Clotho-finetuned release): 0.280 SPIDeR-FL in our harness vs. 0.295 SPIDeR published for this configuration [9] and 0.291 SPIDeR-FL measured for the released base checkpoint with the same `aac-metrics` toolkit [17] — a ≈ 1.1 pp shortfall (plausibly checkpoint-selection or decoding differences) that we disclose; H1 is anchored on CNN14 and unaffected. **Qwen2.5-Omni-7B**: audio-only, talker disabled, greedy decoding, bf16. **SALMONN-13B**: Whisper + BEATs $\rightarrow$ window-level Q-Former $\rightarrow$ Vicuna-13B, fp16, beam 4 (its authored default). **AF3**: Whisper-style encoder + Qwen2-7B decoder, fp32 (its native precision), greedy. Each LALM uses its authors’ recommended decoding; all decode settings are recorded per row.

C. Polyphony Split (RQ2)

We derive pseudo-strong labels with PANNs `Cnn14_DecisionLevelMax` [7] (framewise output, empirically 100 fps). A clip is *polyphonic* iff two AudioSet classes are simultaneously active for ≥ 1 s at confidence $\geq \tau$, evaluated as the longest co-activation run of the same class pair. The project’s public definition fixed $\tau=0.50$; a fallback rule, committed *before* any subset results were seen, required ≥ 300 clips per bucket and otherwise selected the largest stored τ satisfying it. At $\tau=0.50$ only 106 clips were polyphonic (609 clips had no activation at all), so the rule selected $\tau=0.25$: **336 polyphonic / 709 monophonic**. As a validity check, the five reference captions of SED-polyphonic clips mention more distinct sound entities than monophonic ones (mean 4.5 vs. 3.8; ≥ 2 entities in 97% vs. 87% of clips).

D. CHAIR-audio (RQ3)

Vision CHAIR [2] counts caption objects absent from the image against a fixed 80-object list. We port this to audio with the 527-label AudioSet vocabulary (604 surface forms after splitting comma alternatives and dropping six generic structural forms; deterministic word-boundary matching with plural/-ing variants; no learned NLP). An entity in a model caption is *hallucinated* iff it appears neither in (a) the union of the five reference captions nor (b) the clip’s SED tag set at threshold τ (dual criterion). We report CHAIR-s (fraction of captions with ≥ 1 hallucinated entity) at $\tau \in \{0.20, 0.25, 0.30\}$ per the preregistered sensitivity bracket, with matcher coverage.

E. Preregistered Hypotheses and Statistics

Hypotheses were preregistered with BCa bootstrap [18] ($n=1000$ resamples, seed 42) and Holm–Bonferroni correction: **H1**: SPIDeR-FL(AF3) > 0.261 (the published CNN14 baseline; minimum detectable effect 1.04 pp); **H2**: Δ SPIDeR-FL(poly–mono) > 0 per LALM (MDE 1.50 pp); **H3**:

TABLE I
FULL CLOTHO-EVAL (1,045 CLIPS). PUBLISHED SPIDeR-FL ANCHORS:
CNN14 0.261 [8], ENCLAP-BASE 0.291 [17].

Model	SPIDeR-FL	CIDEr-D	SPICE	METEOR	FER $\downarrow$
AST (tag template)	0.068	0.110	0.056	0.095	0.180
Qwen2.5-Omni-7B	0.188	0.286	0.091	0.141	0.005
SALMONN-13B	0.225	0.347	0.110	0.152	0.017
CNN14+BART (DCASE’23)	0.259	0.416	0.118	0.176	0.029
EnCLAP-base	0.280	0.443	0.123	0.179	0.013
Audio Flamingo 3	0.297	0.460	0.137	0.186	0.011

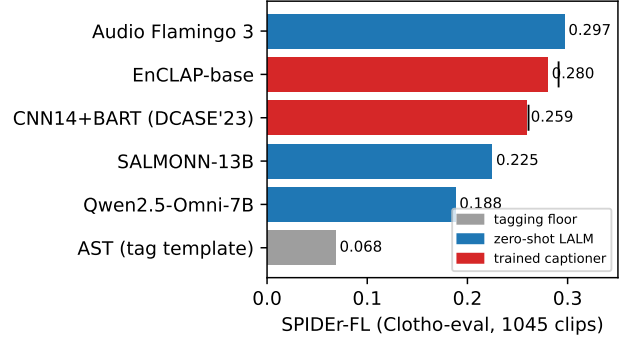


Fig. 1. SPIDeR-FL on full Clotho-eval. Black ticks: published SPIDeR-FL of the trained captioners (CNN14 0.261, EnCLAP-base 0.291).

AF3 $>$ SALMONN (SPIDeR-FL); **H4**: CHAIR(AF3) $<$ CHAIR(SALMONN). The preregistration was drafted before the LALM runs but never formally frozen; we therefore treat it as a declared analysis plan and disclose every deviation (Sec. V-A) [19]. Corpus SPIDeR-FL equals the mean of per-clip scores exactly, so the bootstrap tests the reported metric itself.

IV. RESULTS

A. RQ1: Zero-Shot LALMs vs. Trained Captioners

Table I and Fig. 1 give the main comparison. The ordering is $AST \ll Qwen2.5-Omni < SALMONN < CNN14 < EnCLAP < AF3$ ($0.068 \rightarrow 0.188 \rightarrow 0.225 \rightarrow 0.259 \rightarrow 0.280 \rightarrow 0.297$ SPIDeR-FL). Three observations. (1) Every LALM clears the tagging floor by 3–4 $\times$: LALMs describe scenes rather than list events. (2) Two of three LALMs trail *both* trained captioners, so “zero-shot LALMs beat trained systems” is false in general. (3) The audio-specialist AF3 beats both trained captioners *on every metric* within the harness, zero-shot; its 0.297 also exceeds EnCLAP’s *published* SPIDeR-FL of 0.291 [17], though not EnCLAP’s published CIDEr (0.460 vs. 0.461–0.463 [9]). **H1 is supported**: mean SPIDeR-FL 0.2968, one-sided 95% BCa lower bound 0.2828 > 0.261 + MDE (0.2714), $p \approx 0.001$, significant after Holm correction ($\alpha'=0.0167$). All LALMs are highly fluent (FER ≤ 0.017 ; Qwen2.5-Omni’s 0.005 is the lowest in the table), so their differences reflect content, not disfluency. **H3 is supported**: per-clip paired AF3–SALMONN difference +0.072 (CI lower +0.058, $p \approx 0.001$).

TABLE II
SPIDER-FL ON THE POLYPHONIC VS. MONOPHONIC SUBSETS ($\tau=0.25$;
336/709 CLIPS).

Model	poly	mono	$\Delta(\text{poly}-\text{mono})$
AST (tag template)	0.081	0.062	+0.019
Qwen2.5-Omni-7B	0.250	0.157	+0.094
SALMONN-13B	0.262	0.204	+0.059
CNN14+BART (DCASE'23)	0.298	0.238	+0.060
EnCLAP-base	0.329	0.256	+0.073
Audio Flamingo 3	0.356	0.268	+0.088

TABLE III
CHAIR-AUDIO CAPTION-LEVEL HALLUCINATION RATES (DUAL
CRITERION), LOWER IS BETTER.

Model	CHAIR-s at τ			coverage ($\tau=0.25$)
	0.20	0.25	0.30	
AST (tag template)	0.952	0.956	0.968	1.00
Qwen2.5-Omni-7B	0.534	0.550	0.561	0.91
SALMONN-13B	0.322	0.332	0.337	0.80
CNN14+BART (DCASE'23)	0.343	0.350	0.358	0.75
EnCLAP-base	0.338	0.351	0.357	0.76
Audio Flamingo 3	0.331	0.347	0.355	0.82

B. RQ2: Polyphonic vs. Monophonic

Table II: *all six* systems score higher on the polyphonic subset. **H2 is supported for every LALM** — AF3 $\Delta=+0.088$ (CI lower +0.058), Qwen2.5-Omni +0.094 (+0.071), SALMONN +0.059 (+0.032); all $p \approx 0.001$ (AF3’s Δ , the Holm family member, passes at $\alpha'=0.05$). Because the trained baselines shift by comparable amounts (+0.060 to +0.073), this is a *subset-difficulty* effect rather than an LALM property: event-rich polyphonic clips give captions more matchable content, while the monophonic bucket concentrates quiet, ambiguous recordings (219 clips have no SED activation at $\tau=0.25$ at all) whose descriptions legitimately diverge from references. Notably, no LALM shows a polyphony *penalty* relative to the trained systems at the caption-overlap level — complementing the event-level degradation with polyphony reported by Harish and Abeßer [5]: counting and naming all events gets harder with polyphony, but producing a reference-like *description* does not. The audio-grounded MACE metric [16] (reference implementation, MS-CLAP backend) independently corroborates the direction on the LALMs: poly–mono ΔMACE is +0.028 (AF3), +0.021 (SALMONN), and +0.032 (Qwen2.5-Omni).

C. RQ3: Hallucination

Table III: at $\tau=0.25$, AF3 (0.347), SALMONN (0.332), CNN14 (0.350) and EnCLAP (0.351) cluster tightly, while Qwen2.5-Omni hallucinates markedly more (0.550) and AST’s 0.956 is the expected validity check — an indiscriminate top-5 tag list almost always contains an unsupported label. **H4’s null is retained**: AF3 does *not* hallucinate less than SALMONN (paired difference -0.015 at $\tau=0.25$, $p=0.82$; the sign and decision are stable across all three thresholds, so

the outcome is not threshold-sensitive). The best captioner by overlap metrics is thus not the most grounded one — higher SPIDER-FL is partly earned by richer, more specific captions that carry more hallucination surface. Notably, Harish and Abeßer [5] found Qwen2.5-Omni the *more precise* (less tag-hallucinating) of their LALMs in constrained event tagging; its high CHAIR here suggests hallucination emerges in unconstrained caption generation rather than in the model’s audio grounding — consistent with its competitive MACE. MACE, which scores captions against the audio itself, agrees with the ranking split: AF3 leads (0.593–0.620), but Qwen2.5-Omni (≈ 0.53 –0.56) draws level with SALMONN despite trailing it by 0.037 SPIDER-FL — reference overlap and audio grounding rank models differently. A concrete failure mode: greedy decoding drove Qwen2.5-Omni into a 515-word repetition loop (“tapped, tapped, ...”) on one clip — caught by the fluency-error metric and visible only because per-clip outputs are retained.

V. DISCUSSION AND LIMITATIONS

What “zero-shot” does and does not mean. Zero-shot here denotes the evaluation protocol: no task-specific fine-tuning by us and no in-context examples. It does not mean Clotho-free training: AF3’s training corpus lists the Clotho development pairs (19,195 audio–text pairs) [14], and SALMONN’s pre-training and instruction tuning both include Clotho [13]; Qwen2.5-Omni’s data mix is undisclosed [12]. The evaluation split is standard held-out data for all systems, but prior exposure to the Clotho caption register plausibly contributes to AF3’s overlap-metric advantage — one reason its captions read Clotho-style — and is the same development data the trained baselines were fitted on, so the comparison remains symmetric at the data level. Harish and Abeßer [5] sidestep this by evaluating on synthetic USM soundscapes; our audio-grounded metrics (CHAIR, MACE) serve the same role here. **Metric scope.** Overlap metrics reward reference-like phrasing; AF3’s Clotho-style captions fit them well. CHAIR-audio and MACE add orthogonal, audio-grounded axes on which AF3’s advantage shrinks or disappears. All three are needed for a fair picture. (MACE caveat: its MS-CLAP backend randomly crops a 7 s window per call, giving run-to-run variation of ≈ 0.002 .) **Polyphony split.** SED pseudo-labels inherit PANNs’ biases; AudioSet ontology parents (e.g. “Music”/“Musical instrument”) can co-fire and inflate polyphony. The reference-entity audit supports the split’s direction. The AST cross-check sees only the first 10.24 s. **CHAIR matcher.** The closed vocabulary misses synonyms outside AudioSet surface forms (coverage ≈ 0.8); this bias applies uniformly to all models and largely cancels in the paired H4 comparison. **Compute.** LALMs ran on an A100; baselines on CPU. Only the compute location differs — data, scorer, and references are identical.

A. Deviations from the Preregistration

(1) The preregistration was drafted but never formally frozen (`freeze_date: null`); we disclose rather than claim confirmatory status. (2) RQ2’s metric is $\Delta\text{SPIDER-FL}$

per the preregistration; a project-guide reference to “ Δ MACE” was superseded (and its MACE description was inaccurate). (3) H3/H4 were preregistered on an AudioCaps-single-event set that was never acquired; we substitute paired per-clip comparisons on Clotho-eval, which are statistically stronger. (4) A planned “CLEAN” subset was never operationally defined; H1 uses the full 1,045-clip set — the same set anchoring the 0.261 threshold. (5) The polyphony detector is PANNs SED rather than the announced PaSST, because the definition requires frame-level output; the threshold fallback $\tau=0.50 \rightarrow 0.25$ followed the pre-committed rule. (6) H2’s “paired” wording is undefined for disjoint subsets; we use a two-sample bootstrap. (7) The preregistration also listed three descriptive-only probes (temporal ordering, out-of-distribution clips, a silence negative control); they were cut for scope and never run — no inferential hypothesis depends on them.

VI. CONCLUSION

Whether a zero-shot LALM beats trained audio captioners depends on the model: a current audio-specialist (AF3) does, significantly and on every overlap metric in-harness (and above EnCLAP’s published SPIDeR-FL), while a general omnimodel and an older audio LALM do not. The advantage does not extend to grounding: AF3 hallucinates no less than SALMONN, and caption-overlap gains on polyphonic clips are a dataset effect shared by all systems. Zero-shot captioning quality and audio grounding are separate axes — future evaluation should report both.

AI TRANSPARENCY STATEMENT

Per course requirements: the coding assistant Claude Code (Anthropic; Claude Opus 4.x and Claude Fable 5 models) was used throughout, under continuous human direction and review, for (i) pipeline, scoring, and analysis code, (ii) data analysis and visualization scripts, (iii) literature triage with primary-source verification, and (iv) LaTeX drafting and editing of this paper. Prompts ranged from implementation instructions with explicit verification requirements to review and audit requests; the repository’s session-dated logbook documents the work and the decisions taken under this assistance. All reported numbers are produced by committed scripts operating on raw model outputs; citations, APIs, and checkpoints were verified against primary sources (one published anchor was corrected during a final audit); all analysis decisions (thresholds, fallback rules, deviations) were fixed before results were inspected where noted, and are documented in the repository’s logbook and preregistration files.

CODE AND DATA AVAILABILITY

Code, configs, run manifests, result files, and the project logbook: <https://github.com/Zuraiz270/audio-to-text-captioning-using-LALMs>. Clotho v2.1 is publicly available; model checkpoints are cited in the repository.

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